# Seeing Through the Blur:
# Machine Learning Restoration of Lensed Accretion Disk Images & Black Hole Spin Estimation

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## Abstract

Strong gravitational lensing near black holes produces characteristic photon rings and asymmetric brightness patterns that encode the black hole spin ($a$) and the observer inclination ($i$). In real observations, telescope point spread function (PSF) blurring and detector noise obscure these features. Traditional deconvolution is slow and ill-posed, so we propose a machine learning approach that learns structural spatial priors from data. We will build an end-to-end pipeline that restores sharp accretion disk images from blurred, noisy inputs and simultaneously estimates $a$ and $i$.

Because high-fidelity 3D simulations are too slow for this project, we will synthesize about 1,500 clean 2D intensity profiles with a semi-analytical model. It combines Novikov-Thorne thermal disk emission (bounded by the spin-dependent innermost stable circular orbit, $r_{\mathrm{ISCO}}$), relativistic Doppler boosting, gravitational redshift, and analytical lensing deflection, sampled over $a \in [0, 0.99]$ and $i \in [10°, 80°]$. We will degrade these images with `scipy.ndimage`, convolving with Gaussian PSFs of varying width and adding Gaussian noise, then package the results into paired PyTorch DataLoaders with two targets: the pristine image and the parameters $a$ and $i$.

Our primary model is a dual-headed convolutional autoencoder (U-Net) whose decoder reconstructs the sharp image and whose dense regression head, attached to the bottleneck, predicts $a$ and $i$. We will train it with the Adam optimizer on a joint loss combining image reconstruction error (MSE) and parameter error. On a held-out test set, we will evaluate restoration with the Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM), and parameter inference with $R^2$ and mean absolute error, across a range of noise levels. We will also produce comparison grids of the degraded input, restored image, ground truth, and residual map. If time allows, we will compare against a conditional GAN or diffusion model paired with a separate CNN regressor.

Because the training data are synthetic, we will also discuss limitations in transferring the results to real telescope observations.

## Team Roles:

- Aaron: Project management and physics lead. Manages and documents the synthetic dataset and validates its physical consistency.
- Rafael: Model development. Designs and trains the dual-headed U-Net and its joint reconstruction and parameter-estimation loss.
- Deniz: Degradation pipeline and evaluation. Builds the PSF and noise pipeline and DataLoaders, then computes restoration and parameter-inference metrics and produces the comparison figures and presentation.
- Alex: Final paper and integration. Writes the manuscript and implements a classical deconvolution baseline for comparison.

All members will present their own component and contribute the corresponding section of the final paper.